



Cookie Monster Secret Recipe

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Easy

Web Exploitation

picoCTF 2025

browser_webshell_solvable

AUTHOR: BRHANE GIDAY AND PRINCE NIYONSHUTI N.

Description

Cookie Monster has hidden his top-secret cookie recipe somewhere on his website. As an aspiring cookie detective, your mission is to uncover this delectable secret. Can you outsmart Cookie Monster and find the hidden recipe? Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.

Its current status is:

NOT_RUNNING

Launch Instance

Hints

1 2 3

Cookies aren't just for eating - they're also used in web technologies!

42,995 users solved



94%

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Author: The Analyst: Hyposelenia

Challenge: Cookie Monster Secret Recipe

Category: Web Exploitation

Date: 02/21/26



I. Objective

The objective of this challenge is to analyze how cookies are used in a web application and determine whether modifying or decoding cookie values can grant access to restricted information, specifically the hidden flag.

II. Background

Cookies are small pieces of data stored in a user's browser that websites use to maintain sessions, preferences, or authentication states. In web exploitation challenges, insecure handling of cookies may allow attackers to manipulate or decode sensitive values to bypass access controls or retrieve confidential information.

This challenge hints that the secret information may be stored client-side within browser cookies.

III. Tool Used

- **Web Browser** - Used to access and interact with the challenge website
- **Browser Developer Tools (Inspect)** - Used to view page structure and cookies
- **CyberChef** - Used to decode encoded cookie values

IV. Methodology

1. Launched the challenge instance provided by picoCTF.



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Its current status is:

NOT_RUNNING

Launch Instance

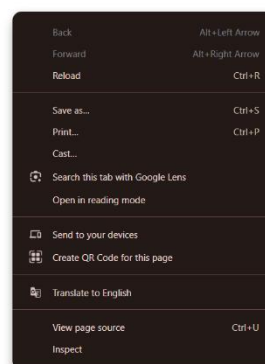
2. Accessed the challenge web URL using a web browser

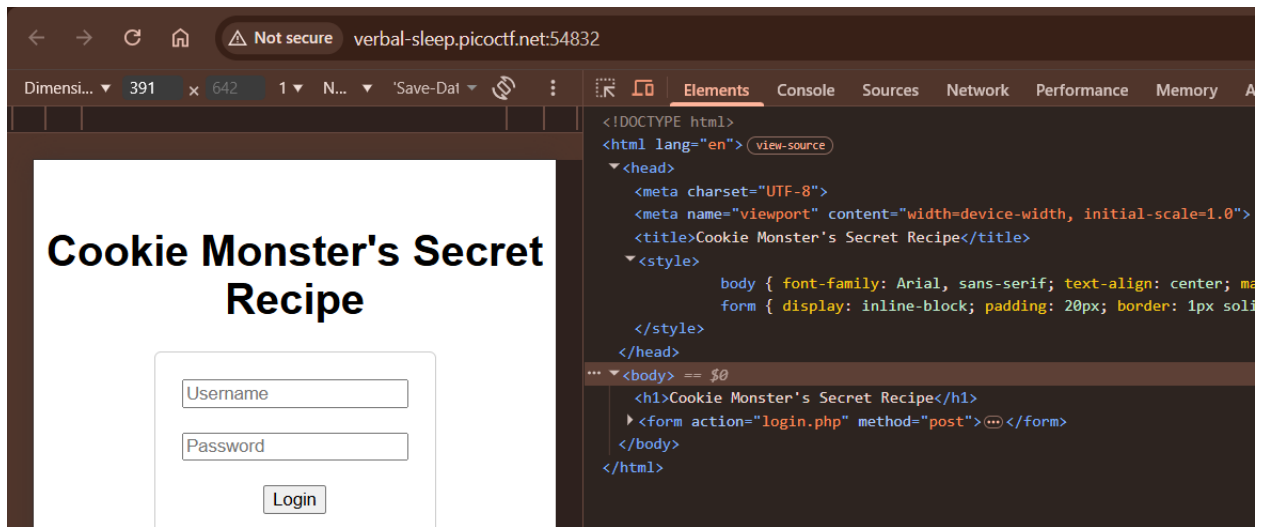
verbal-sleep.picocftf.net:54832

Cookie Monster's Secret Recipe

3. **Inspected** the landing page and analyzed its structure; it appeared normal with no visible vulnerabilities.

Cookie Monster's Secret Recipe

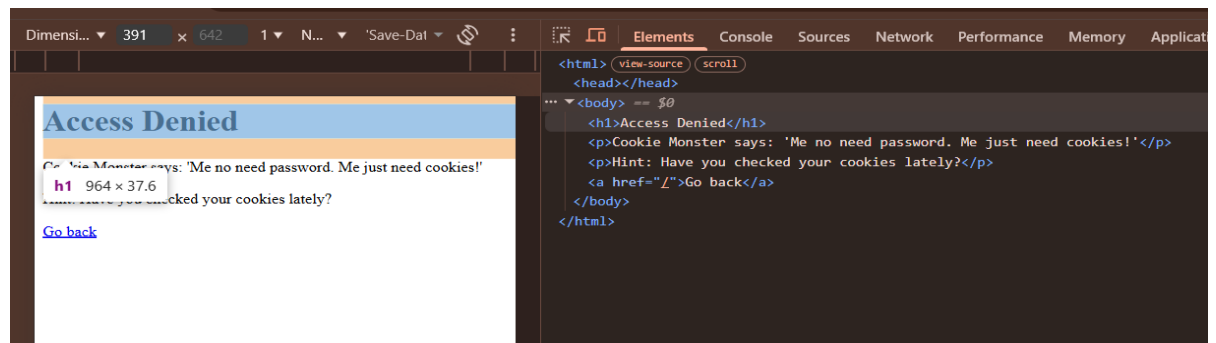




4. Attempted to log in using a made-up credentials.



5. After logging in, I was redirected to a page displaying “Access Denied.”

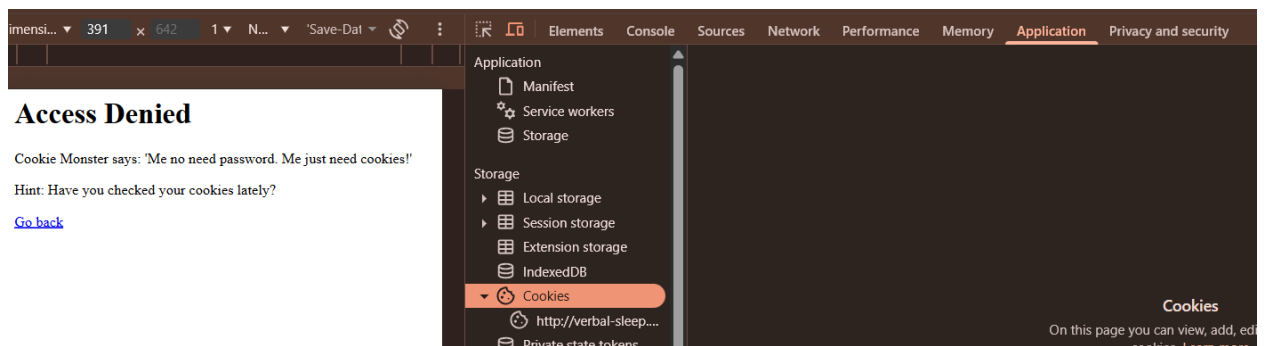




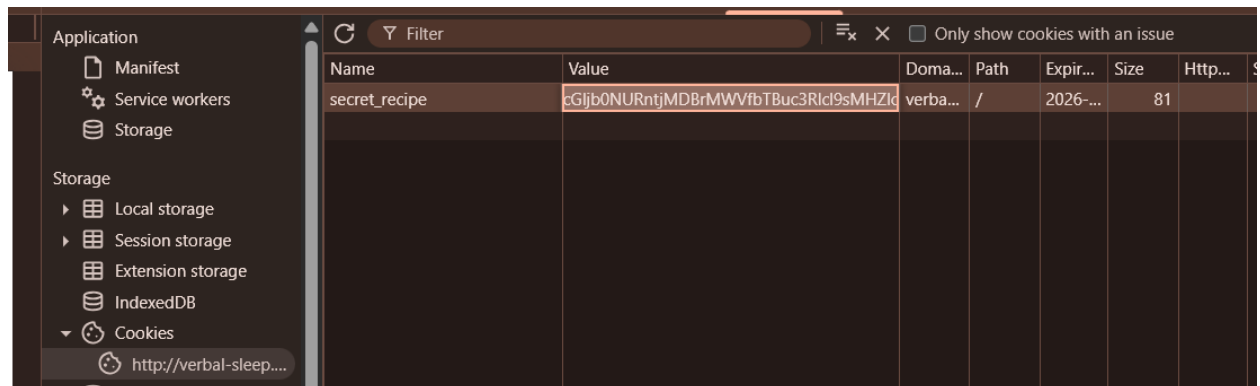
However, a hint on the page suggested checking browser cookies.

```
<p>Cookie Monster says: 'Me no need password. Me just need cookies! '</p>
<p>Hint: Have you checked your cookies lately?</p>
<a href="/>Go back</a>
```

6. Navigated to **Application Tab** → **Cookies**.



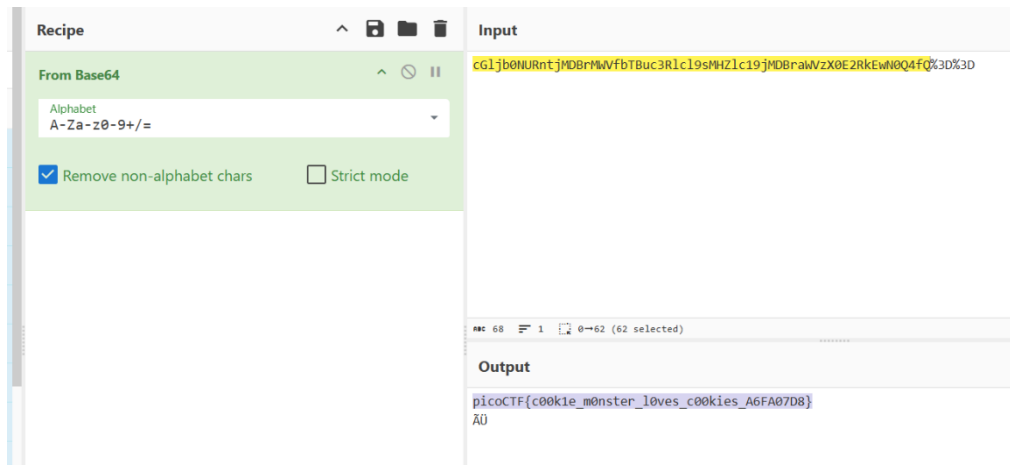
7. Identified a cookie named **secret_recipe** whose value appeared encoded due to the mix of letters and numbers.



8. Copied the cookie value and pasted it into **CyberChef**.



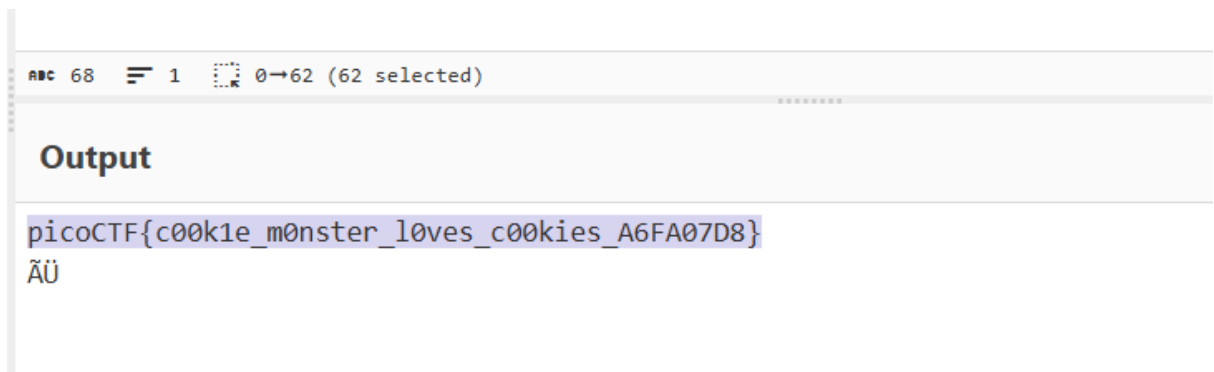
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9. Applied the **“From Base64”** operation, which successfully decoded the value.
10. The decoded output revealed the flag.

V. Result

picoCTF{c00k1e_m0nster_l0ves_c00kies_A6FA07D8}





VI. Explanation

The challenge exploited improper storage of sensitive data within browser cookies. Instead of securely validating access on the server side, the application stored the secret information in a client-side cookie encoded in Base64. Since Base64 is an encoding method and not encryption, the data could be easily decoded using publicly available tools such as CyberChef.

This allowed the flag to be retrieved without bypassing authentication mechanisms.

VII. Conclusion

This challenge demonstrates the importance of secure cookie handling in web applications. Sensitive information should never be stored client-side without proper encryption and server-side validation. Through careful inspection and decoding of cookies, the flag was obtained without advanced exploitation techniques, highlighting a common web security weakness.

— **The Analyst: Hyposelenia**